

Akhila Katukam

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PROFESSIONAL SUMMARY

Versatile **Data Engineer and AI/ML practitioner** with 3.5+ years of experience designing and implementing **ETL/ELT pipelines** for structured and unstructured data across healthcare and financial services. Skilled in **Python, SQL, Spark, Airflow, dbt, and Snowflake**, with hands-on experience in **multi-cloud platforms (AWS, GCP, Azure)**. Skilled in **advanced SQL (window functions, query tuning, schema design)**, Python, Spark, Airflow, dbt, and Snowflake. Proven ability to build **real-time streaming pipelines (Kafka, Kinesis, Event Hubs)**, integrate data from APIs and CDC tools (Fivetran, Debezium), and deliver **governed, high-quality datasets**. Experienced in **data modeling (star/snowflake)**, **observability (CloudWatch, Datadog, Prometheus)**, and compliance frameworks (**HIPAA, SOX, GDPR, CCPA, SOC 2**). Strong background in **MLOps (MLflow, Feast, Databricks Feature Store)**, enabling predictive analytics and scalable ML workflows.

TECHNICAL SKILLS

Programming & Languages	Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, spaCy), SQL, Terraform, CloudFormation, C++ (Familiarity), Java (Familiarity), Scala (Exposure), Go (Exposure), R (Exposure)
Cloud Platforms	AWS (S3, Redshift, Glue, Lambda, SageMaker, EMR, Kinesis), GCP (BigQuery, Vertex AI, Pub/Sub, Dataflow), Azure (Databricks, Synapse, Delta Lake, ADLS Gen2, Unity Catalog, Event Hubs), Apache Beam, IaC (Terraform, CloudFormation)
ETL & Data Engineering	Apache Airflow, Spark, Kafka, dbt, Informatica, Cloud-native ETL tools (Glue, Dataflow, ADF), ERD tools, SSIS, Talend, Fivetran, Prefect, Dagster, Flink, CDC Pipelines (Debezium, Kafka, Fivetran)
Databases & Warehousing	Snowflake, PostgreSQL, MySQL, Oracle, SQL Server, Cosmos DB, MongoDB (NoSQL), Cassandra, Hive
Visualization & BI Tools	Tableau, Power BI, Looker, Sigma
MLOps & DevOps	Docker, Git, Jenkins, CI/CD pipelines, Kubernetes (AKS, EKS, GKE), Terraform, ARM Templates, Feast, Databricks Feature Store, MLflow, Monitoring & Observability (CloudWatch, Datadog, Prometheus)

Machine Learning	Feature Engineering, ML-ready datasets, NLP, Deep Learning, Model Deployment
Special Knowledge	Data Architecture, Data Modeling (Star/Snowflake, Semantic Layers), Data Quality Validation, Governance & Compliance (HIPAA, SOX, FINRA, GDPR, CCPA, SOC 2, IAM Policies, Data Lineage)

PROFESSIONAL EXPERIENCE

Data Engineer | T-Mobile

Dallas, TX | March 2024 – Present

- Designed and maintained **10+ ETL/ELT pipelines** (Airflow, Glue, dbt) ingesting 100K+ healthcare records/day across AWS/GCP/Azure, ensuring HIPAA/SOX/FINRA compliance and improving scalability with **Databricks and Synapse**.
- Built **5+ Snowflake data marts** and optimized **BigQuery datasets** with partitioning/clustering, reducing costs by 25% and cutting dashboard refresh times by 40%.
- Developed **real-time streaming pipelines** with **Kafka, Spark, and exposure to Kinesis/Event Hubs**, processing 50K+ transactions/hour and reducing fraud detection latency by 30%.
- Optimized BigQuery and Snowflake queries using **partitioning, clustering, and window functions**, reducing query runtime by 40%.
- Implemented **CDC ingestion (Fivetran, Debezium)** to integrate SaaS, API, and database sources; automated schema validation and data quality checks with **Great Expectations-like frameworks**.
- Automated monitoring and incident response with **AWS Lambda, CloudWatch, and Datadog**, cutting manual oversight by 40% and boosting uptime.
- Partnered with data scientists to deliver **ML-ready datasets** and managed lifecycle with **MLflow and Feast Feature Store**, improving predictive model accuracy by 15%.
- Developed **observability datasets** to track data lineage, cost, and pipeline SLAs; enhanced governance with **GDPR, CCPA, SOC 2** compliance.
- Delivered **CI/CD workflows** via Jenkins, Git, and Terraform; **orchestrated** hybrid cloud deployments with Kubernetes and Docker.
- Collaborated in Agile sprints, ensuring 95% of deliverables aligned with business outcomes.

Tech Stack: Airflow, Glue, dbt, Kafka, Spark, Snowflake, BigQuery, Synapse, Databricks, MLflow, Feast, Terraform, Jenkins, Kubernetes, Docker, CloudWatch, Datadog, AWS/GCP/Azure

Junior Data Engineer | Broadridge

Hyderabad, India | Aug 2021 – June 2023

- Assisted in building **3+ real-time streaming pipelines** with Kafka/Spark and exposure to **Debezium/Fivetran**

for CDC, enabling 50K+ financial transactions/hour with anomaly detection.

- Contributed to migration of **20+ ETL workflows** into **AWS Redshift, dbt, and Informatica**, improving scalability and cutting costs by 20%.
- Designed **SQL-based marts** to accelerate compliance reporting turnaround by 35%.
- Optimized large-scale SQL transformations, improving query performance by 25% on analytical workloads.
- Delivered **5+ Tableau dashboards** for compliance teams, improving visibility into high-value transactions and fraud detection.
- Optimized **SQL** transformations and complex joins on large-scale analytical workloads, improving query performance by 25%.
- Built scalable **ELT/ETL workflows** with validation and **data quality frameworks** to reduce inconsistencies and improve trust in reporting.
- Applied **data governance practices** (HIPAA, SOX, GDPR, CCPA) to strengthen compliance and regulatory readiness.
- Supported Agile sprints with 100% Jira task completion; collaborated with cross-functional teams on analytics and reporting needs.

Tech Stack: Kafka, Spark, Redshift, dbt, Informatica, SQL, Tableau, Airflow, CDC (Debezium, Fivetran), Agile, Jira

PROJECTS

Prediction of Contaminants in Water Using Machine Learning

- Built classification and regression models to predict water contaminants using physicochemical data from WHO, EPA, and Kaggle.
- Engineered features and performed EDA using Pandas, Matplotlib, and Seaborn to uncover high-risk indicators.
- Applied Random Forest, XGBoost, and Neural Networks with GridSearchCV to achieve 92%+ accuracy on unseen data.
- Deployed model via Flask API to deliver real-time predictions; monitored logs and performance metrics.
- Engineered data ingestion and preprocessing pipelines; deployed model via Flask API for real-time predictions.

Fake News Detection using Natural Language Processing

- Preprocessed textual data using tokenization, stop word removal, stemming, and TF-IDF vectorization for feature extraction.
- Trained and evaluated Naive Bayes and Logistic Regression models to classify fake vs. real news with high precision.

- Conducted performance evaluation using confusion matrix, accuracy, and F1-score to compare models.
- Achieved 90%+ accuracy; demonstrated NLP's effectiveness in identifying misinformation patterns in online news.

CERTIFICATIONS

- **Databricks Lakehouse Fundamentals**
- **Azure Data Fundamentals (DP-900)**
- **Certified Associate in Python Programming**

EDUCATION

Master of Science in Information Technology

St. Francis College – New York, NY | May 2025

Bachelor of Technology in Computer Science

Geethanjali College of Engineering and Technology – India | 2023